
Linear Programming

CS5491: Artificial Intelligence

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Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course



ANNOUNCEMENTS



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- ❑ **No Class Next Week -- Happy Chinese New Year!**



WHAT TO EAT?

We are trying healthy by finding the optimal amount of food to purchase.

We can choose the amount of **stir-fry** (ounce) and **boba** (fluid ounces).

Health Goals

- $2000 \leq \text{Calories} \leq 2500$
- $\text{Sugar} \leq 100g$
- $\text{Calcium} \geq 700mg$

Food	Cost	Calories	Sugar	Calcium
stir-fry (per oz)	1	100	3	20
Boba (per fl oz)	0.5	50	4	70

What is the cheapest way to stay "healthy" with this menu?

How much **stir-fry** (ounce) and **boba** (fluid ounces) should we buy?



CAN SEARCH ALGORITHMS HELP?

Essence of search algorithms:

- explore all nodes in search tree until you reach the one you want
- use additional information to avoid search all nodes

Can we adopt a similar strategy here?



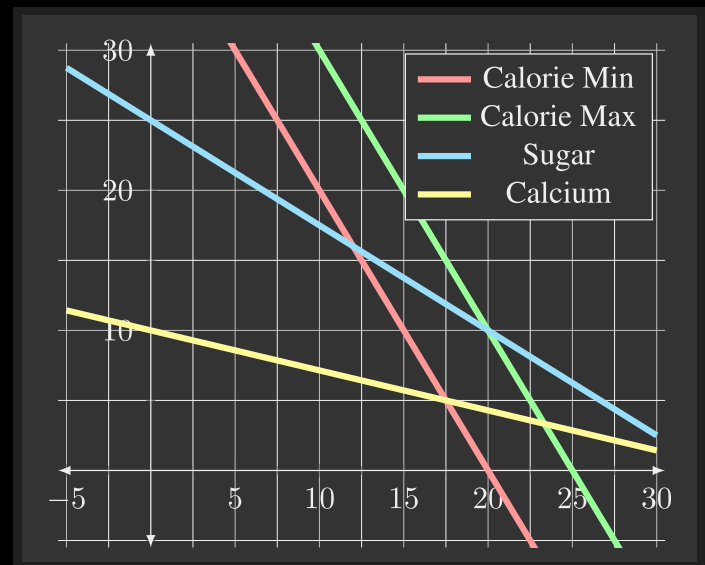
OPTIMIZATION

Problem Description

Optimization Representation

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

Graphical Representation



WHAT TO EAT?: CONSTRAINTS

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CONSTRAINT SATISFACTION PROBLEMS

Map Coloring

Any $\mathbf{x}$

s.t. $\mathbf{x}$ satisfies constraints



WHAT TO EAT?: COSTS

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Diet Problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \text{cost}(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{x} \text{ satisfies constraints} \end{aligned}$$



OPTIMIZATION FORMULATION

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Diet Problem

$$\begin{aligned} \min_{\mathbf{x}} \quad & \text{cost}(\mathbf{x}) \\ \text{s.t.} \quad & \text{calories}(\mathbf{x}) \leq \text{limit} \\ & \text{calories}(\mathbf{x}) \geq \text{limit} \\ & \text{sugar}(\mathbf{x}) \leq \text{limit} \\ & \text{calcium}(\mathbf{x}) \geq \text{limit} \end{aligned}$$

OPTIMIZATION FORMULATION

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Diet Problem

$$\begin{array}{ll}\min_{\mathbf{x}} & 1x_1 + 0.5x_2 \\s.t. & 100x_1 + 50x_2 \geq 2000 \\ & 100x_1 + 50x_2 \leq 2500 \\ & 3x_1 + 4x_2 \leq 100 \\ & 20x_1 + 70x_2 \geq 700\end{array}$$



OPTIMIZATION FORMULATION

Diet Problem

$$\begin{array}{ll}\min_{\mathbf{x}} & c_1x_1 + c_2x_2 \\ \text{s.t.} & a_{1,1}x_1 + a_{1,2}x_2 \geq b_1 \\ & a_{2,1}x_1 + a_{2,2}x_2 \leq b_2 \\ & a_{3,1}x_1 + a_{3,2}x_2 \leq b_3 \\ & a_{4,1}x_1 + a_{4,2}x_2 \geq b_4\end{array}$$

$$A = \begin{bmatrix} 100 & 50 \\ 100 & 50 \\ 3 & 4 \\ 20 & 70 \end{bmatrix} \quad b = \begin{bmatrix} 2000 \\ 2500 \\ 100 \\ 700 \end{bmatrix} \quad \text{and} \quad c = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

OPTIMIZATION FORMULATION

Diet Problem

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & a_{1,1}x_1 + a_{1,2}x_2 \geq b_1 \\ & a_{2,1}x_1 + a_{2,2}x_2 \leq b_2 \\ & a_{3,1}x_1 + a_{3,2}x_2 \leq b_3 \\ & a_{4,1}x_1 + a_{4,2}x_2 \geq b_4\end{array}$$

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OPTIMIZATION FORMULATION

Diet Problem

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & -a_{1,1}x_1 - a_{1,2}x_2 \leq -b_1 \\ & a_{2,1}x_1 + a_{2,2}x_2 \leq b_2 \\ & a_{3,1}x_1 + a_{3,2}x_2 \leq b_3 \\ & -a_{4,1}x_1 - a_{4,2}x_2 \leq -b_4\end{array}$$

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OPTIMIZATION FORMULATION

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$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{A} \mathbf{x} \leq \mathbf{b}\end{array}$$

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LINEAR PROGRAMMING

Linear objective with linear constraints

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{A}\mathbf{x} \leq \mathbf{b}\end{array}$$

As opposed to general optimization

$$\begin{array}{ll}\min_{\mathbf{x}} & f_0(\mathbf{x}) \\ s.t. & f_i(\mathbf{x}) \leq 0, i = 1, \dots, M \\ & \mathbf{a}_i^T \mathbf{x} = \mathbf{b}_i, i = 1, \dots, P\end{array}$$



LINEAR PROGRAMMING

Different formulations

Inequality Form

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b}\end{array}$$

General Form

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} + d \\ s.t. & \mathbf{Gx} \leq \mathbf{h} \\ & \mathbf{Ax} = \mathbf{b}\end{array}$$

Standard Form

$$\begin{array}{ll}\min_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$



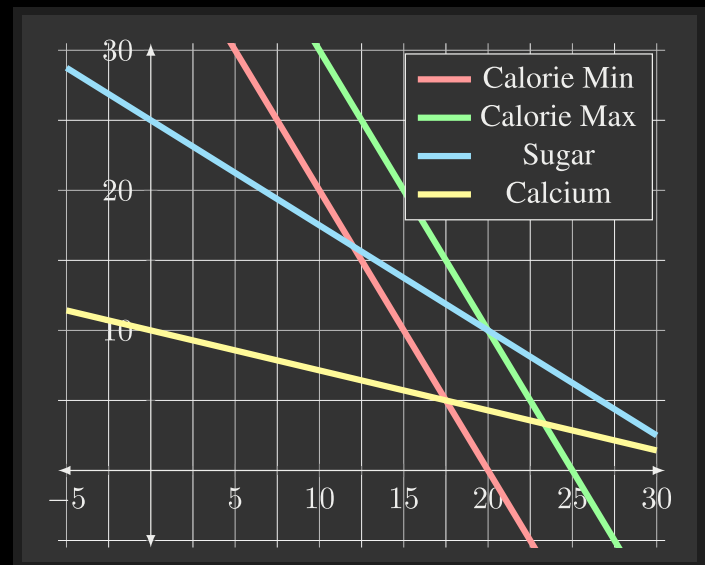
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GRAPHICS REPRESENTATION

What shape does this inequality represent?

$$a_1x_1 + a_2x_2 \leq b_1$$



COST CONTOURS

Given the cost vector $[\mathbf{c}_1, \mathbf{c}_2]^T$ where will,

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LP GRAPHICAL REPRESENTATION

Inequality Form

$$\begin{array}{ll}\min & \mathbf{c}^T \mathbf{x} \\ s.t. & \mathbf{Ax} \leq \mathbf{b}\end{array}$$



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Start with no constraints



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Start with no constraints

Add one constraint at a time



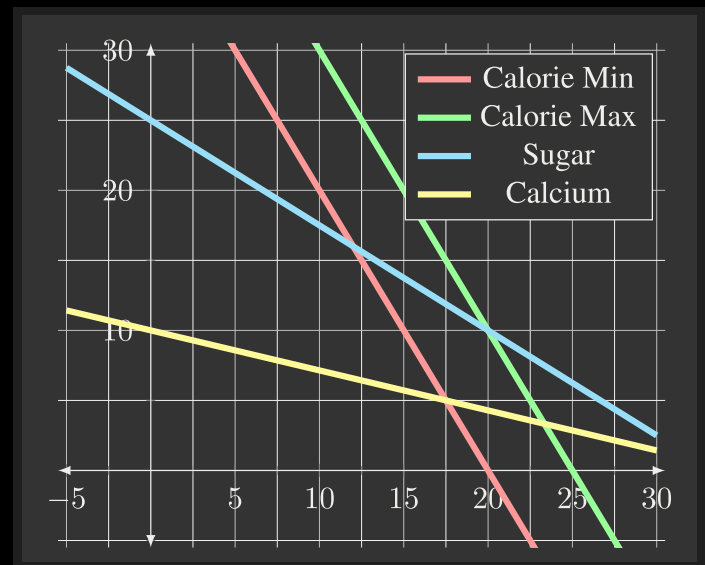
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SOLVING AN LP

Solutions are at feasible intersections of constraint boundaries!!

